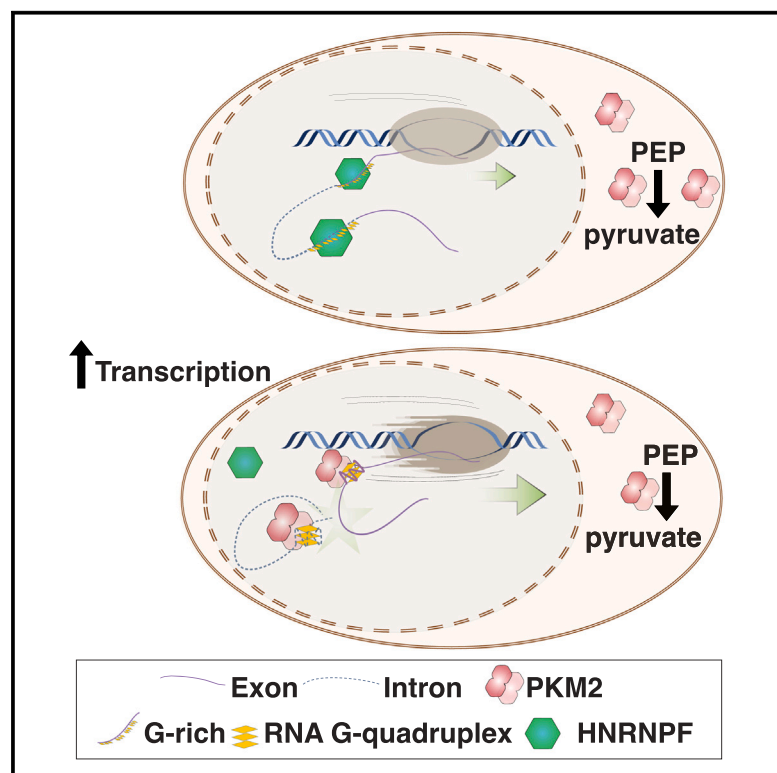


Nuclear PKM2 binds pre-mRNA at folded G-quadruplexes and reveals their gene regulatory role

Graphical abstract



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In brief

Anastasakis et al. find that PKM2 binds G-quadruplexes (rG4) forming on precursor mRNAs. They show that the balance of folded and unfolded rG4s controls transcriptional output of rG4-containing RNAs (rG4ome). The rG4ome encodes EMT components, and the removal of PKM2 from the nucleus results in reduced tumor growth and metastasis.

Highlights

- Nuclear PKM2 binds folded RNA G4s on pre-mRNA
- G4 folding during transcription promotes mRNA expression
- G4-containing RNAs encode EMT components
- Nuclear PKM2 promotes lung metastasis



Article

Nuclear PKM2 binds pre-mRNA at folded G-quadruplexes and reveals their gene regulatory role

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SUMMARY

Nuclear localization of the metabolic enzyme PKM2 is widely observed in various cancer types. We identify nuclear PKM2 as a non-canonical RNA-binding protein (RBP) that specifically interacts with folded RNA G-quadruplex (rG4) structures in precursor mRNAs (pre-mRNAs). PKM2 occupancy at rG4s prevents the binding of repressive RBPs, such as HNRNPF, and promotes the expression of rG4-containing pre-mRNAs (the “rG4ome”). We observe an upregulation of the rG4ome during epithelial-to-mesenchymal transition and a negative correlation of rG4 abundance with patient survival in different cancer types. By preventing the nuclear accumulation of PKM2, we could repress the rG4ome in triple-negative breast cancer cells and reduce migration and invasion of cancer cells *in vitro* and in xenograft mouse models. Our data suggest that the balance of folded and unfolded rG4s controlled by RBPs impacts gene expression during tumor progression.

INTRODUCTION

Pyruvate kinases are enzymes that catalyze the final step of glycolysis by generating pyruvate and ATP from phosphoenolpyruvate and ADP. In mammals, pyruvate kinases are encoded by two genes: *PKLR*, expressed in liver and red blood cells, and *PKM*, expressed in all other tissues.¹ *PKM* produces two isozymes, PKM1 and PKM2, that differ by 22 amino acids.² Proliferating cells and tumor cells typically express PKM2, which supports their shift in energy metabolism toward aerobic glycolysis (“Warburg effect”).^{3–5} A tightly regulated alternative splicing event produces the two PKM isoforms by inclusion of either exon 9 in PKM1 or exon 10 in PKM2. In proliferating cells, a set of MYC-controlled HNRNP proteins bind sequences flanking exon 9 to favor PKM2 formation.⁶ This MYC-driven response results in high expression levels of PKM2 in tumors.^{4,7}

Accumulating evidence shows that non-metabolic activities of PKM2 in tumors promote aerobic glycolysis, cell proliferation, migration, and survival.^{4,7–10} Some of the PKM2 non-metabolic activities may be linked to the nuclear translocation of PKM2,^{11,12} which is widely observed in cancers and contributes

to tumor progression and aggressiveness.¹³ ERK1/2-dependent phosphorylation of serine 37 (S37) and O-GlcNAcylation appear to drive PKM2 nuclear entry.^{14–16} In the nucleus, PKM2 has been hypothesized to participate in gene regulation through unknown molecular mechanisms.^{11,12,17,18}

RNA interactome capture, as well as crosslinking and immunoprecipitation (iCLIP) approaches recently revealed that PKM2 can directly bind RNA *in vivo*,^{19–22} analogous to >20 other metabolic enzymes.^{23–25} The functional role of these “moonlighting” RNA-binding proteins (RBPs) is poorly understood. Modulation of their enzymatic activity through RNA ligands was reported for GAPDH and ENO1.^{26,27} Some may post-transcriptionally regulate their targets, as observed for the iron response protein 1 (IRP1) that directly interacts with ferritin messenger RNAs (mRNAs) to regulate iron metabolism.²⁸

Our data reveal that PKM2, but not PKM1, is an RBP that selectively and specifically interacts with RNA G-quadruplex (rG4) structures with high affinity.²⁹ Of the various RNA structures with important functional roles,³⁰ rG4s—formed by two or more stacked guanine tetrads built by Hoogsteen hydrogen bonding—are among the most stable.^{29,31} Intense efforts were



directed toward identifying rG4s and interacting RBPs to understand their impact on gene regulation in healthy and diseased cells.^{32–38} *In vitro* and *in silico* approaches revealed over 13,000 sites in the human transcriptome with the potential to form rG4s.^{39–41} There is scattered evidence that rG4s may influence many aspects of posttranscriptional regulation, including alternative polyadenylation, splicing, and miRNA biogenesis.^{32,35,42–45} Some rG4 sequence elements in 5' untranslated regions (UTRs) and possibly 3' UTRs of mRNA have been reported to negatively affect translation.^{34,46,47} Nevertheless, the existence and relevance of rG4s *in vivo* remain the subject of controversial discussion.⁴⁸ While immunofluorescence experiments with a specific antibody detected potentially folded rG4s in human cells,⁴⁹ chemical mapping—focusing on mature, predominantly cytoplasmic mRNA—suggested they are globally unfolded.⁴⁰ We and others identified members of the essential machinery⁴⁰ resolving rG4s comprising cytoplasmic RNA helicases^{50,51} and RBPs.^{52,53} The conservation of thousands of rG4-forming sequences across the vertebrate transcriptome⁵⁴ suggests that folded rG4s are important for gene expression. However, the cellular and developmental context in which these rG4s globally fold has yet to be captured. We present evidence that rG4 structures form in precursor mRNAs (pre-mRNAs) to promote their expression and identify RBPs that stabilize or unfold these structures to regulate gene expression.

We observed that nuclear PKM2 bound thousands of putative rG4-forming sequences (PRG4Ss) within pre-mRNAs. Furthermore, nuclear PKM2 increased the expression of rG4-containing mRNAs through competition with repressive RBPs, such as the heterogeneous ribonucleoprotein F (HNRNPF), which bound the same unfolded G-rich sequences. Our data suggest that the interplay between stabilizing and repressive RBPs at rG4s regulate gene expression and could be the key to understanding the effect of PKM2 in promoting tumorigenesis. Furthermore, we observed an upregulation of rG4-containing mRNAs (the “rG4ome”) at hybrid epithelial/mesenchymal states of breast cancer cells undergoing epithelial-to-mesenchymal transition (EMT). Consistently, exclusion of PKM2 from the nucleus resulted in a reduced rG4ome and reduced migration and invasion of triple-negative breast cancer (TNBC) cells *in vitro* and in xenograft models. Taken together, we propose that rG4s represent a cancer-regulon and identify PKM2-rG4 ribonucleoproteins (RNPs) as an attractive targetable vulnerability to control tumor progression.

RESULTS

In vitro-selected RNAs binding PKM2 are enriched for rG4s

Previous studies suggest that PKM2 may bind RNA *in vivo*.^{19–22} To identify sequence and/or structural determinants of PKM2 RNA-binding in an unbiased manner, we performed RNA Bind-n-seq (RBNS) assays.⁵⁵ Libraries of fully randomized RNA oligonucleotides (29-mers) were mixed with recombinant PKM2 expressed either in *E. coli* or purified from HEK293 cells (Figures S1A and S1B). The isolated PKM2-bound small RNAs were quantified by next-generation sequencing. The enrichment of sequence elements was calculated and compared with the input library composition (Figure 1A). We observed that RNAs

bound by PKM2 showed a modest enrichment for 5-mers that contained G-triplets (Figure 1B). The G-triplet-containing sequences resembled the building blocks of canonical G-quadruplex (rG4) structures, which comprise three stacked Hoogsteen-bonded guanine tetrads connected by linkers of variable length²⁹ (“three-tetrad” rG4; ((GGG(N_{1–7}))₃GGG)).^{56,57} To examine a potential association of PKM2 with G4-forming RNAs, we calculated the enrichment of sequences that contained repetitions of any of the 64 NNN triplets separated by linkers. Indeed, sequences matching the criteria for canonical rG4s were substantially enriched (Figure 1C). The enrichment of rG4-forming sequences was eliminated when performing RBNS in the presence of Li⁺ rather than K⁺ (Figure 1D), indicating interaction of PKM2 with folded rG4 structures, which typically require large monovalent cations (K⁺ or Na⁺) for their formation.^{29,58}

Next, we validated the interaction of PKM2 with folded G4 structures using electrophoretic mobility shift assays (EMSAs). PKM2 bound the RNA and DNA form of the widely used G4-forming sequence found in the promoter region of *MYC* with similar affinity ($K_D \sim 300$ nM) as the G4-melting helicase DHX36 ($K_D \sim 100$ nM, Figures 1E and S1C).^{51,59} PKM2 did not bind the same sequence modified with 8-aza-7-deazaguanine, which prevents Hoogsteen base-pairing and therefore G4 structure formation (Figure 1F). Moreover, in EMSAs, PKM2 showed minimal interaction with less stable “two-tetrad” rG4 structures (Figure S1D), suggesting that it requires stably folded three-layer rG4 structures for binding and likely cannot act as an rG4 chaperone inducing their folding.⁶⁰

We next asked whether rG4 binding by PKM2 affects its enzymatic activity or whether it melts rG4 structures *in vitro* as many other rG4 binders do.^{50,51} Circular dichroism spectroscopy revealed that PKM2 did not melt folded rG4s (Figure S1E). *In vitro* enzymatic assays showed that PKM2-rG4 complexes had comparable enzymatic activities to free PKM2 in both less active, as well as more active states induced by the physiological activator fructose-1,6-bisphosphate (FBP). This suggested that PKM2's substrate and FBP binding sites are not occluded by RNA-binding (Figure S1F), consistent with interactome studies in mouse embryonic stem cells⁶¹ that find RNA crosslinking at amino acids distant from substrate and FBP binding sites (Figure S2A). Taken together, our results indicate that PKM2 selectively and specifically interacts with folded rG4s with high affinity and does not alter their structure.

HNRNPF binds unstructured rG4 sequences in contrast to PKM2

We wanted to identify additional RBPs with rG4 binding activity like PKM2 and reanalyzed RBNS datasets for 78 distinct RBPs deposited in the ENCODE database.⁶² Among all tested proteins, only HNRNPF and the closely related HNRNPH2 exhibited appreciable selectivity for the canonical, rG4-forming motif ((GGG(N_{1–7}))₃GGG) (Figure 1G). We performed our own RBNS experiments in the presence of K⁺ or Li⁺ to identify whether HNRNPF preferred folded or unfolded rG4s as a substrate. In contrast to PKM2, we found that HNRNPF bound rG4-forming sequences (containing four GGG repeats) with higher affinity in the presence of Li⁺ (Figures 1H and 1I). This indicated that HNRNPF does bind tightly to GGG repeats via its three RNA

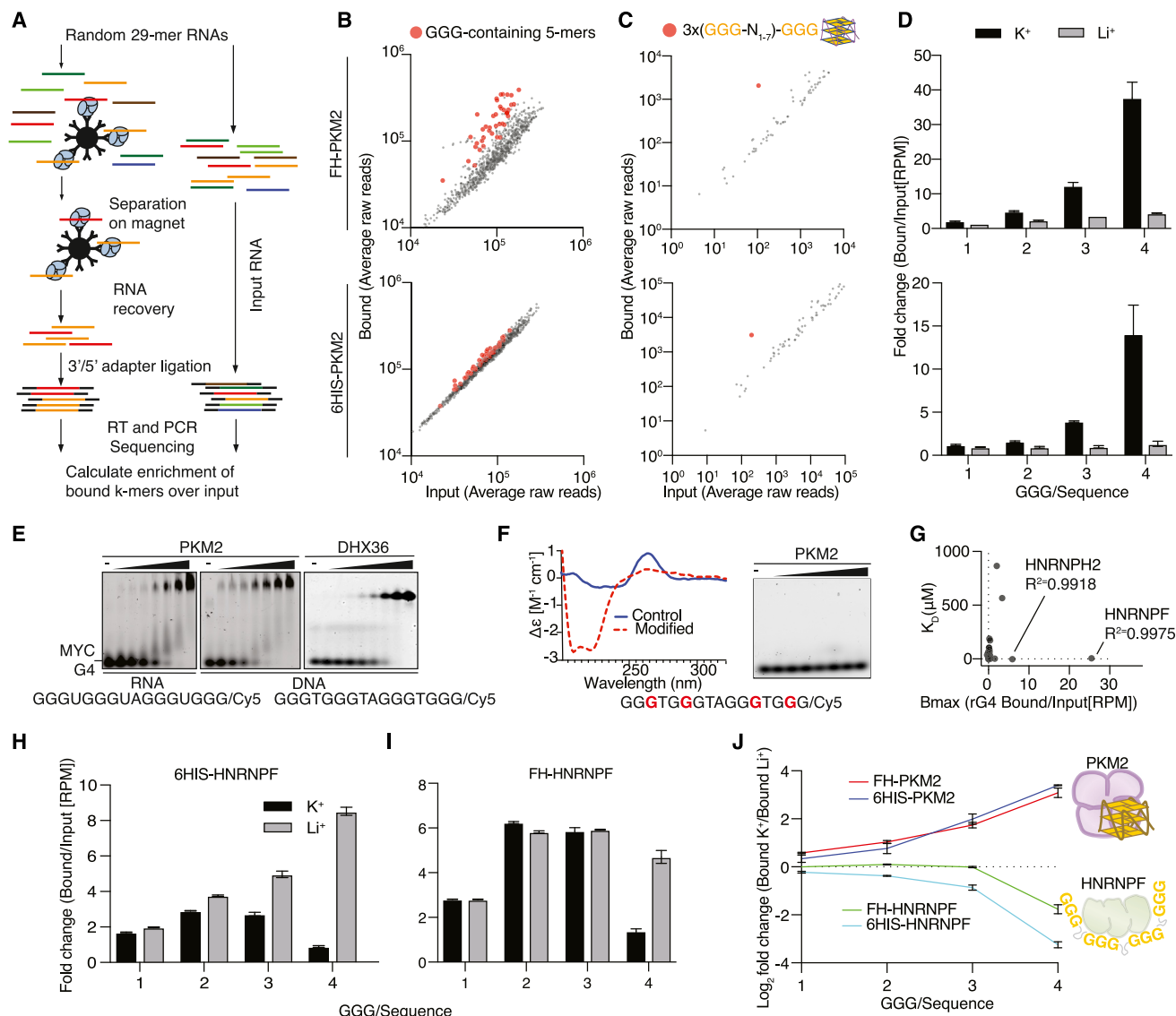


Figure 1. PKM2 binds folded and HNRNPF binds unfolded rG4s *in vitro*

(A) Schematic of the RBNS approach.
(B and C) Scatterplot of relative abundance of 5-mers (B) or four repeats of any nucleotide triplet separated by any 1–7 nucleotides (XYZ(N₁₋₇)₃XYZ) in (C) input and PKM2-bound RBNS libraries.
(D) Enrichment of sequences containing 1–4 G-triplets in RBNS performed in the presence of K⁺ or Li⁺.
(E) EMSA using 6HIS-PKM2 or DHX36 with the G4-forming RNA or DNA 16-mer derived from the MYC promoter.
(F) Circular dichroism (CD) spectrum of the oligonucleotide from (D) containing 8-Aza-7-deazaguanine (left). Modified bases are indicated in red (lower). EMSA using 6HIS-PKM2 and the modified sequence (right).
(G) Affinity (K_D) versus maximal enrichment (B_{max}) of reanalyzed, publicly available RBNS data. R² from fit to $Y = B_{max} \times X/(K_d + X)$.
(H and I) Enrichment of sequences containing 1–4 G-triplets in 6HIS-HNRNPF (H) or FH-HNRNPF (I) RBNS performed in the presence of K⁺ or Li⁺.
(J) PKM2 prefers binding to folded rG4 elements, while HNRNPF binds linear sequences. Ratio of enrichment (log₂ bound over input) of sequences containing 1–4 G-triplets from RBNS experiments in the presence of K⁺ over Li⁺ for FH and 6HIS tagged PKM2 and HNRNPF. Error bars represent standard error of mean (SEM).

recognition motif (RRM) domains^{63–65} but unlike PKM2 cannot bind structured rG4s (Figure 1J). While our approach did not reveal any additional proteins binding folded rG4 structures, we nevertheless identified HNRNPF as an RBP that could potentially compete with PKM2 for access to rG4-forming sequences *in vivo*. This observation guided our following functional analysis

of the possible gene regulatory impact of the PKM2-rG4 interaction by considering the role of HNRNPF as a competitor.

Nuclear PKM2 and HNRNPF bind PRG4Ss in pre-mRNAs

Previous iCLIP experiments studying the role of PKM2 in translational regulation in the cytoplasm did not identify specific

sequence interactions but rather showed it contacted the ribosome A-site.²⁰ Therefore, we focused on nuclear RBP-RNA interactions to determine whether PKM2 and HNRNPF interacted with and potentially competed for rG4-forming sequences *in vivo*.

We comprehensively mapped the RNA interactome of PKM2 and characterized its RNA recognition elements using fluorescent photoactive ribonucleoside-enhanced crosslinking and immunoprecipitation (fPAR-CLIP).^{66,67} Altogether, we performed 21 fPAR-CLIP experiments with endogenous or FLAG/HA (FH)-tagged PKM2 and HNRNPF in HEK293 and MDA-MB-231 cells after biochemical fractionation of nucleus and cytoplasm (Table S1). MDA-MB-231 cells express high levels of phosphorylated PKM2 (PKM2pS37), which accumulates in the nucleus.¹³ We also verified PKM2 presence in the nucleus of HEK293 cells by probing with PKM2pS37 antibody (Figures S2B–S2D). The tagged versions of the proteins were expressed at significantly lower levels than the endogenous protein (6- to 20-fold), mitigating concerns about experimental artifacts introduced by overexpression (Figures S3A and S3B). We also performed two biological replicates of HiTS-CLIP^{67,68}—which does not rely on photoreactive ribonucleosides but rather uses short-wavelength UV for RNP crosslinking—for endogenous nuclear and cytoplasmic PKM2. For all experiments, fluorescent imaging of the crosslinked and ribonuclease-treated immunoprecipitates reveals a single band migrating at ~100 and ~75 kDa, corresponding to the PKM2-RNP and HNRNPF-RNP, respectively, ligated to a fluorescent 3' adapter oligonucleotide (Figure S3C). Cross-linked PKM2 and HNRNPF RNPs were isolated, and protein-bound RNA fragments were transformed into small RNA cDNA libraries for next-generation sequencing. We defined RBP binding sites by generating clusters of overlapping sequence reads that harbor T-to-C conversions diagnostic of 4-thiouridine (4SU)-crosslinking at higher frequencies than expected by chance⁶⁹ (Table S1).

Biological replicates of individual fPAR-CLIP experiments showed high correlation (R^2 between 0.8 and 1) when comparing the aligned crosslinked reads using deeptools.⁷⁰ Interestingly, nuclear PKM2 and HNRNPF fPAR-CLIP experiments clustered closely across all cell lines and separate from cytoplasmic PKM2 (Figure 2A). Consequently, binding sites from nuclear and cytoplasmic PKM2 fPAR-CLIPs were largely non-overlapping (Figure S3D). Our binding sites reflected the cellular compartment of origin (Figure 2B), and nuclear PKM2 and HNRNPF mainly bound pre-mRNAs (Figure 2B; Table S1).

The number of crosslinked reads per transcript in fPAR-CLIP showed little correlation to transcript abundance (Figure S3E), suggesting that our experiments captured sequence- or structure-dependent determinants of binding rather than unspecific interactions. In line with our RBNS experiments and previous studies, a motif analysis revealed a strong enrichment of G-rich sequences in HNRNPF and nuclear PKM2 binding sites (Figures 2C, S3F, and S3G). In the cytoplasm, PKM2 binding sites showed no enrichment of sequence elements, strongly suggesting that the RNA contacts were indeed mediated by interactions with the ribosome.²⁰

Our *in vitro* experiments showed PKM2 and HNRNPF binding at folded and unfolded rG4s, respectively. Consistently, our

fPAR-CLIP experiments identified strong binding of PKM2 and HNRNPF to PRG4Ss, a feature we did not observe for other nuclear RBPs in fPAR-CLIP experiments (Figure 2D). We used EMSA to validate PKM2 binding to two different PRG4Ss that we discovered in our PAR-CLIP data, confirming that they could fold into rG4s (Figures 2E and S4A–S4D). Recombinant PKM2 phosphorylated at S37 (Figure S4E)—the PKM2 form predominantly found in the nucleus—retained even higher affinity for the same rG4s (Figure 2E). Interestingly, neither recombinant PKM1 nor the closely related *E. coli* homolog pykF (43% identity) bound these PRG4Ss (Figures 2E, S1B, and S1D). This suggests that PKM2 rG4 binding activity arose with the vertebrate-specific splicing of *PKM* and is linked to the 22 amino acids encoded by the alternative exon 10, either directly or indirectly through conformational changes.

We next mined 221 PAR-CLIP experiments available to us (Table S2) to identify other RBPs interacting with PRG4Ss in living cells. Nuclear PKM2 and HNRNPF ranked as the most prominent PRG4S-binding proteins in addition to DHX36 and CNBP—which we previously studied as rG4 resolving proteins—and DDX3X with a broader activity in resolving structured mRNA (Figures 2F and 2G; Table S2). Notably, unlike HNRNPF and PKM2, which are nuclear proteins, DHX36, CNBP, and DDX3X localize mainly to the cytoplasm. Thus, we proceeded to focus on elucidating the relationship between nuclear PKM2 and HNRNPF and their impact on gene expression of rG4-containing mRNAs.

PKM2 and HNRNPF compete for access to rG4s

The evidence gathered so far led us to conclude that when PKM2 localizes to the nucleus, it interacts with folded rG4 structures. Meanwhile, HNRNPF may act as a competitor, as it binds to the same, but unfolded, PRG4Ss. If PKM2 and HNRNPF competed for access to the same PRG4Ss, one would predict that (1) the binding profile on PRG4Ss substantially overlaps and (2) that removing one of the RBPs from the nucleus would result in increased binding of the other to PRG4Ss. We decided to test these predictions in MDA-MB-231 cells. fPAR-CLIP for endogenous PKM2 or HNRNPF (Figures 2B and 2C) showed that both proteins shared a substantial number of binding sites, which were enriched at PRG4Ss (Figures 3A–3C).

Next, we depleted PKM2 from the nucleus and asked whether occupancy of HNRNPF on PRG4Ss increased. To remove PKM2 from the nucleus, we used CRISPR-Cas9-mediated editing to introduce a nuclear export signal (NES) at the PKM1/2 C terminus.⁷¹ As expected, the fusion protein (PKMNES) did not show any trace of nuclear localization (Figure S5A). Then, we compared four biological replicates each of HNRNPF fPAR-CLIP experiments using MDA-MB-231 cells expressing wild-type (WT) PKM or PKMNES and found that excluding PKM2 from the nucleus increased HNRNPF interaction with PRG4Ss (Figure 3D). Thus, PKM2 and HNRNPF indeed compete for the same set of binding sites.

Depletion of nuclear PKM2 decreases abundance of its pre-mRNA targets

Now that we established that PKM2 and HNRNPF preferentially interacted with PRG4Ss found across pre-mRNAs, we

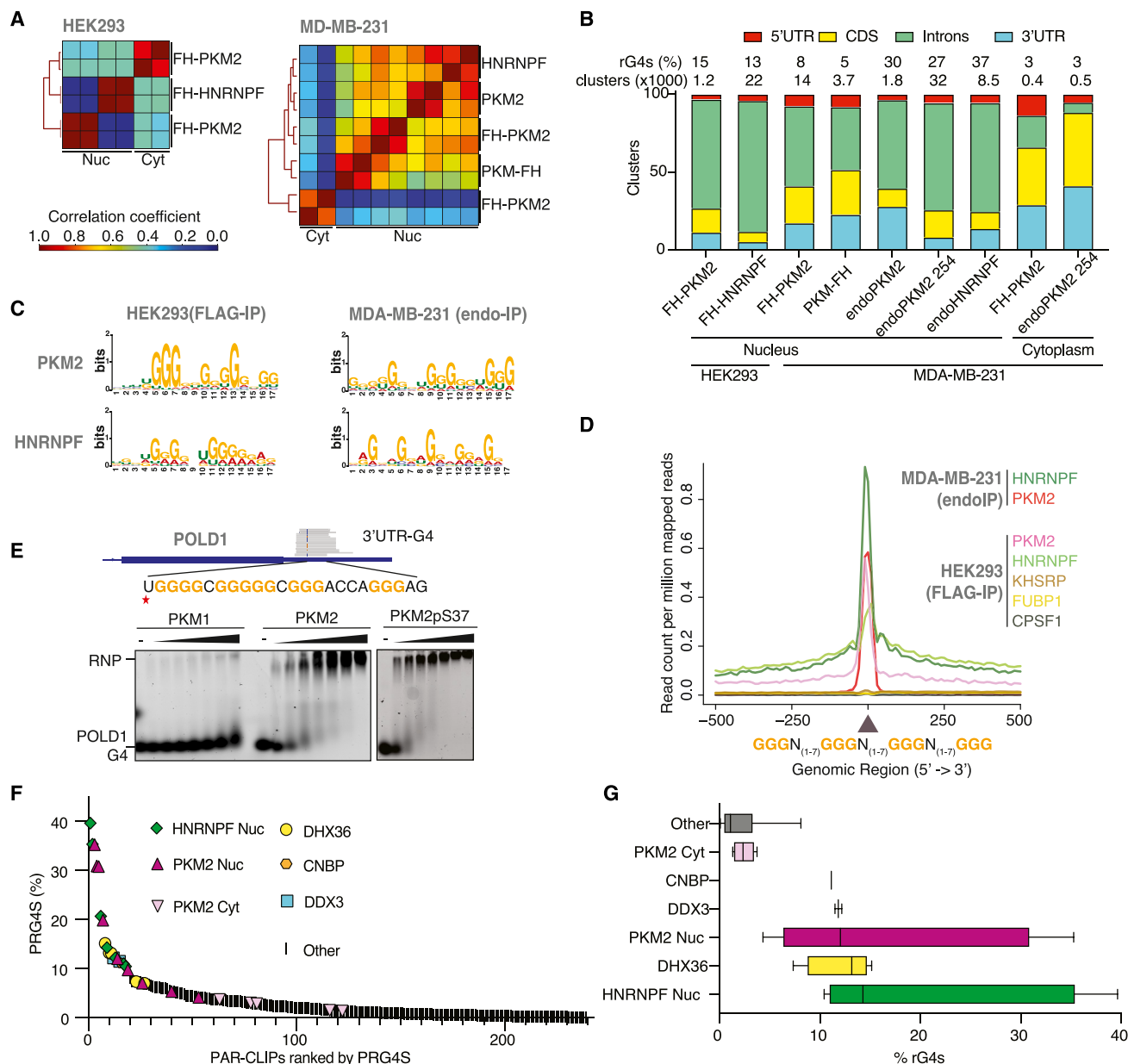


Figure 2. PKM2 and HNRNPF bind to PRG4Ss in vivo

(A) Correlation heatmap across all PKM2 and HNRNPF fPAR-CLIP experiments, separated by cell line. Spearman correlation coefficient indicated.

(B) Average distribution of PKM2 or HNRNPF binding sites across different mRNA annotation categories in the cytoplasm or the nucleus of HEK293 and MDA-MB-231.

(C) MEME motif in nuclear PKM2 and HNRNPF binding sites.

(D) Metagene plot of crosslinked footprints from representative fPAR-CLIP experiments centered on PRG4Ss.

(E) (Upper) Alignment of nuclear FH-PKM2 fPAR-CLIP reads around a PRG4S in the 3' UTR of POLD1 mRNA. (Lower) EMSA using 6HIS-PKM1, 6HIS-PKM2, or 6HIS-PKM2pS37 the rG4-forming oligoribonucleotide corresponding to PKM2 binding site on POLD1.

(F and G) Reanalysis of 233 PAR-CLIP experiments showing the percentage of binding sites per experiment containing PRG4Ss. (G) summarizes replicate experiments as boxplots.

next aimed to understand their gene regulatory impact using loss- and gain-of-function analyses. In our HEK293 model cell lines, PKM2 is the main pyruvate kinase expressed, and thus we did not directly knockout or knockdown PKM2 to avoid antic-

ipated secondary effects due to changes in metabolic activity. Rather, we generated stable HEK293 cell lines expressing FH-PKM2 fused to an NES at its C terminus that abrogated nuclear localization and depleted remaining endogenous PKM2—which

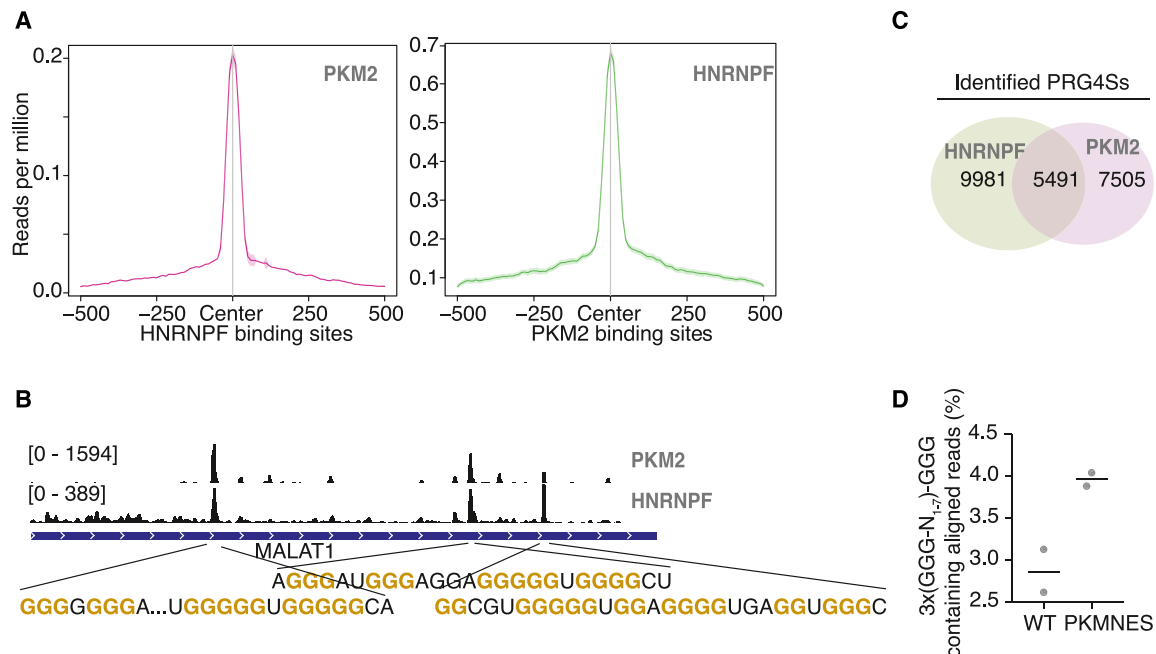


Figure 3. PKM2 and HNRNPF compete for access to PRG4Ss *in vivo*

(A) PKM2 and HNRNPF bind similar sites, shown by metagenome analysis of crosslinked sequence read coverage from PKM2 PAR-CLIP around HNRNPF binding sites (left) or from HNRNPF PAR-CLIPs around PKM2 binding sites. (B) Coverage of HNRNPF and PKM2 binding sites on MALAT1. (C) Venn diagram of binding site overlap for endogenous PKM2 and HNRNPF. (D) Proportion of sequence reads containing the canonical PRG4S in four biological replicates of HNRNPF fPAR-CLIP in MDA-MB-231 expressing wild-type (WT) PKM or PKM fused to a nuclear export signal (PKMNES).

may enter the nucleus—with siRNAs targeting the 3' UTR of the *PKM* mRNA (Figures 4A and S5B). To ensure that tagging at the N-terminal or C-terminal of the protein does not impair its metabolic activity, we measured either pyruvate kinase activity of cell extracts or ATP production by purified proteins in the presence of ADP and phosphoenolpyruvate (Figures S5C and S5D). Tagging of the protein with N-terminal FH and C-terminal V5-tags did not change its activity, while unexpectedly, PKM2-NES constructs exhibited even higher pyruvate kinase activity as measured by both methods. This implies that nuclear shuttling of PKM2 does not impair its catalytic activity.

Using these cell lines for transcriptome profiling, we found that exclusion of PKM2 from the nucleus led to a significant decrease in mRNA levels of nuclear PKM2 PAR-CLIP targets. The magnitude of this effect was dependent on the overall strength of binding (defined by the number of crosslinked reads normalized by transcript abundance) with the expression levels of the top 10% of targets ($n = 208$) reduced on average by 10% (Figure 4B). We further refined our target list by restricting binding sites to those containing the canonical rG4 motif either in exons, introns close (≤ 200 nt) to splice sites, or deep within the intronic sequence space (Figure 4B). In all cases, we observed that loss of nuclear PKM2 resulted in reduced levels of target mRNAs. In exons or close to splice sites, even a single PKM2-bound rG4 was sufficient to elicit this effect, while bound intronic rG4s further removed from splice sites were not as predictive of regulation by PKM2 (Figure 4B). This phenomenon was independent

of the cell line used, and similar results were obtained in another cell line with nuclear PKM2,¹³ MCF-7, after depletion of nuclear PKM2 (Figures S5E–S5G). We concluded that in the nucleus, PKM2 promotes expression of its targets.

Nuclear PKM2 and HNRNPF have opposite effects on the expression of rG4-containing mRNA

We hypothesized that in the nucleus, PKM2 binds to rG4s forming on nascent RNA and protects them from unfolding by RBPs binding linear G-rich elements, such as HNRNPF, resulting in up-regulation of rG4-containing RNAs. Therefore, increasing levels of PKM2 in the nucleus should result in a commensurate increase of expression of rG4-containing mRNA (the rG4ome), while increased levels of HNRNPF should oppose this effect. To test this prediction, we generated a stable HEK293 cell line expressing an inducible PKM2 fused to a nuclear localization signal (FH-PKM2-NLS) (Figure 4C) and quantified mRNA expression using RNA sequencing (RNA-seq). Considering the substantially increased levels of PKM2 in the nucleus, we expected that PKM2 is now able to engage with a substantial proportion of the rG4ome. Therefore, we binned transcripts based on the presence and number of canonical PRG4Ss and found that increasing levels of nuclear PKM2 significantly increased the abundance of rG4-containing mRNAs with the magnitude of the effect depending on the number of predicted rG4s within exons or near the splice sites (Figure 4D). At the same time, increasing the levels of HNRNPF (Figure 4E) resulted in a

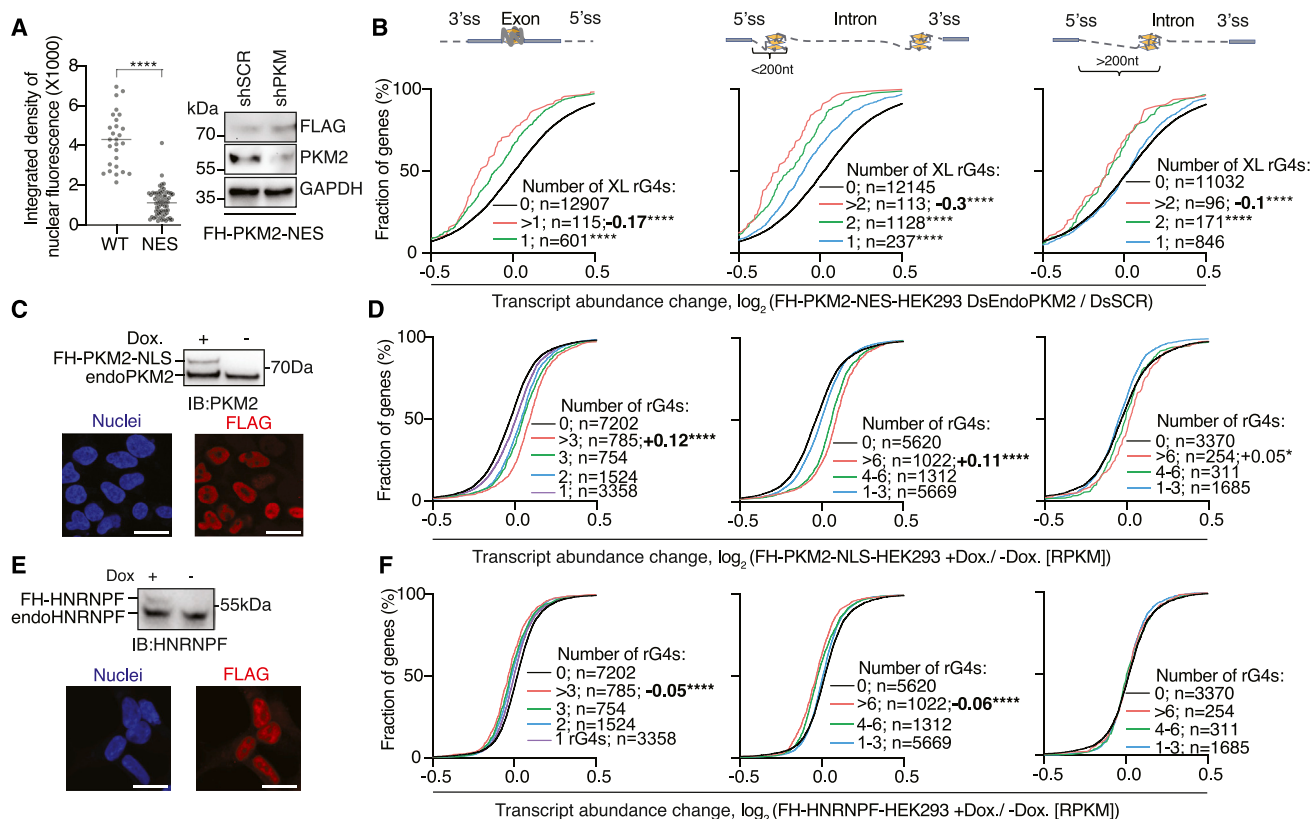


Figure 4. PKM2 and HNRNPf competitively regulate rG4-containing transcripts *in vivo*

(A) (Left) Quantification of nuclear FLAG signal in cells expressing FH-PKM2 and FH-PKM2-NES. (Right) Immunoblot confirming silencing of endogenous PKM2 and expression of FH-PKM2-NES.

(B) mRNA expression changes after silencing of endogenous PKM2 in cells expressing FH-PKM2-NES. The empirical cumulative distribution function (CDF) of nuclear PKM2 targets binned according to the number of canonical rG4s crosslinked to PKM2 (GGG(N₁₋₇)₃-GGG) in their exons (left), in introns within 200 nt from splice sites (middle), or greater than 200 nt (right).

(C) (Upper) Immunoblot confirming expression of FH-PKM2-NLS after doxycycline induction. (Lower) Immunofluorescence analysis showing nuclear localization of FH-PKM2-NLS; scale bars, 15 μ m.

(D) mRNA expression changes after expression of FH-PKM2-NLS. CDF of genes binned according to the number of canonical rG4s (GGG(N₁₋₇)₃-GGG) in their exons (left), in introns within 200 nt from splice sites (middle), or greater than 200 nt (right).

(E) (Upper) Immunoblot confirming expression of FH-HNRNPf after doxycycline induction. (Lower) Immunofluorescence analysis showing nuclear localization of FH-HNRNPf; scale bars, 15 μ m.

(F) mRNA expression changes after expression of FH-HNRNPf. CDF of genes binned according to the number of canonical rG4s (GGG(N₁₋₇)₃-GGG) in their exons (left), in introns within 200 nt from splice sites (middle), or greater than 200 nt (right). Nuclei, Hoechst 33342 (blue); FLAG, Alexa Fluor 594 (red); scale bars, 20 μ m. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$, Mann-Whitney U test (MWU).

decrease of those rG4-containing mRNAs (Figure 4F). Taken together, our results indicate that the balance of folded and unfolded rG4 elements directly affects mRNA expression and that shifting this balance to favor folding—either by removing the repressive HNRNPf or expression of PKM2—will result in increased expression of the rG4ome.

Nuclear PKM2 promotes expression of rG4-containing pre-mRNAs in a co-transcriptional manner

DNA G4 structures forming in gene promoters represent regulatory elements controlling transcription.⁷² Nevertheless, while PKM2 is capable to interact with DNA G4 structures *in vitro* (Figure 1E), nuclear PKM2 did not affect expression of genes with G4s in their promoter in HEK293 or MCF-7 cells (Figures S5G and S5H). Moreover, nuclear PKM2 did not affect the expression

of genes with G4 elements on the template strand of the gene body, i.e., the reverse complement of the RNA transcript, further supporting that rG4 structures, rather than DNA-G4 structures, are the relevant PKM2 targets in living cells (Figure S5I).

Our data showed that PKM2 prominently bound pre-mRNA, i.e., nascent transcripts, which led us to test if this interaction affected transcriptional dynamics. To address this, we used chromatin immunoprecipitation sequencing (ChIP-seq) targeting the unphosphorylated C-terminal domain (CTD) of the large RNA polymerase II (RNA Pol II) subunit RBP1, which mainly recovers inactive RNA Pol II recruited to gene promoters. We found a decreased occupancy of RNA Pol II at transcription initiation sites of PRG4S-containing mRNAs upon expression of FH-PKM2-NLS (Figure 5A). Another set of ChIP-seq experiments targeting CTD phosphorylated at serine 2 (S2P), which marks

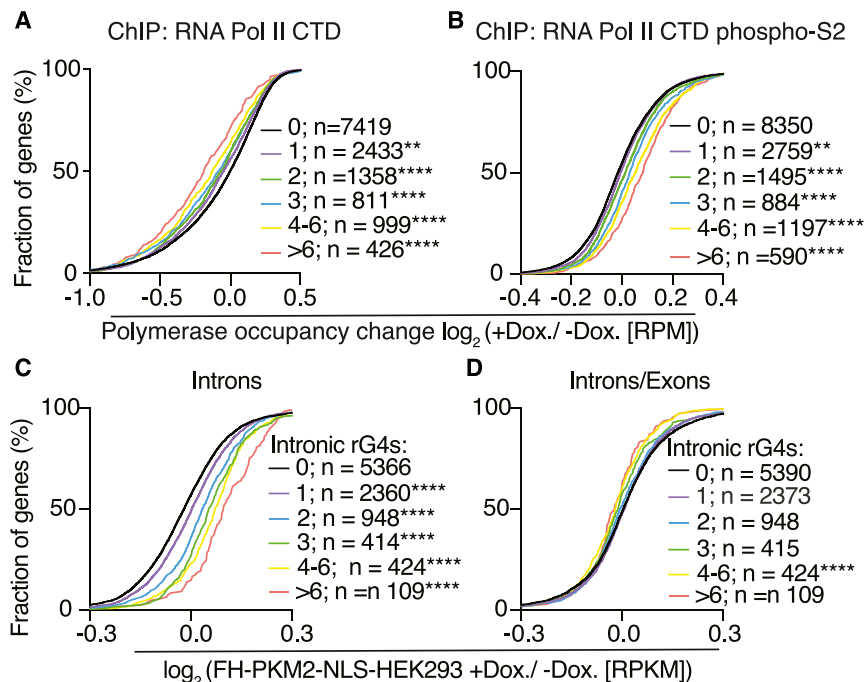


Figure 5. PKM2 binding and rG4 formation promote transcription elongation

(A and B) Polymerase occupancy changes of total RNA Pol II near transcription start site (A) or elongating RNA Pol II (CTD phosphorylated at S2) along the gene body (B). Transcripts were binned according to the number of canonical rG4 motifs (GGG(N₁₋₇)₃GGG) in their exons or within 200 nt of the 5' or 3' splice site.

(C and D) Abundance changes of intronic reads (C) or ratio of intronic/exonic reads (D) upon expression of FH-PKM2-NLS. Transcripts were binned according to the number of canonical rG4 motifs (GGG(N₁₋₇)₃GGG) in their exons or within 200 nt of the 5' or 3' splice site. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$; Mann-Whitney U test (MWU).

elongating RNA Pol II, revealed an increase in elongating RNA Pol II upon expression of FH-PKM2-NLS (Figure 5B) and thus, likely, an increased transcriptional output of PRG4S-containing mRNAs. This is mirrored in an overall increase of RNA-seq sequence reads derived from introns of PRG4S-containing mRNAs (Figure 5C), a metric that can be used to measure transcriptional activity.⁷³ At the same time, the overall ratio of intronic to exonic reads on rG4 mRNAs did not change (Figure 5D), indicating that despite the increase in nascent transcripts, mRNA processing remained efficient. Thus, we concluded that PKM2 affects PRG4S-containing mRNAs during transcription.

rG4-containing mRNAs are upregulated in cells undergoing partial EMT

Previous reports show that nuclear translocation of PKM2 promotes invasive properties of tumor cells.^{17,18} Consistently, a gene set enrichment analysis (GSEA) of transcriptional changes in FH-PKM2-NLS-HEK293 cells showed a significantly increased expression of multiple gene sets connected with tumorigenesis, most notably EMT-related genes (Figure 6A).⁷⁴ To test whether the rG4ome is related to EMT phenomena, we reanalyzed gene expression data from an established breast cancer model of EMT.⁷⁵ In this model, cells can be separated into three phenotypic states: a basic, less tumorigenic state (E), an epithelial/mesenchymal hybrid (E/M) state that is strongly associated with tumor invasiveness and metastasis, and a final mesenchymal state (xM) that is less tumorigenic. We found that the top upregulated mRNAs after overexpression of PKM2 in the nucleus were also upregulated when cells transit from the E to the aggressive E/M state. By contrast, these same mRNAs were downregulated by HNRNPF overexpression (Figure 6B), highlighting a potential competition with PKM2 and supporting a previously reported inhibitory role of HNRNPF on

again (Figure 6E). We speculated that the PKM2-rG4 interaction may allow a rapid modulation of key gene sets marked by rG4 to promote a migratory phenotype, which correlates to transitional phases of tumorigenesis and metastasis, thereby providing the observed link between the rG4ome expression and cancer. Consistently, cell migration regulatory pathways and genes related to cell motility were significantly enriched in a Gene Ontology (GO) analysis of rG4-containing transcripts upregulated by PKM2 in the nucleus (Figure S6).

rG4ome expression correlates with poor patient prognosis in cancer

Considering that EMT genes, cell migratory pathways, and genes related to cell motility are enriched in the rG4ome, we wondered whether rG4ome upregulation is a hallmark of cancer subtypes or aggressive cancers in general. We reanalyzed the entire set of RNA-seq data available from The Cancer Genome Atlas (TCGA) and stratified samples from various cancers by expression levels of the rG4ome (mRNAs with ≥ 4 rG4 elements per transcript). We observed for all cancer subtypes that some samples stood out with comparatively high expression of the rG4ome. Independent of cancer type, high expression of the rG4ome correlated with poor prognosis, e.g., in acute myeloid leukemia, breast cancer, renal adenocarcinoma, or cutaneous melanoma (Figure S7A). The poor prognosis for patients with cancers expressing high levels of the rG4ome was even more pronounced when combining all available TCGA datasets (Figure S7B).

Exclusion of PKM2 from the nucleus reduces lung metastases in mouse xenografts

In a previous study,¹³ we found a small molecule, TEPP-46, which acts as an isoform-specific pyruvate kinase activator

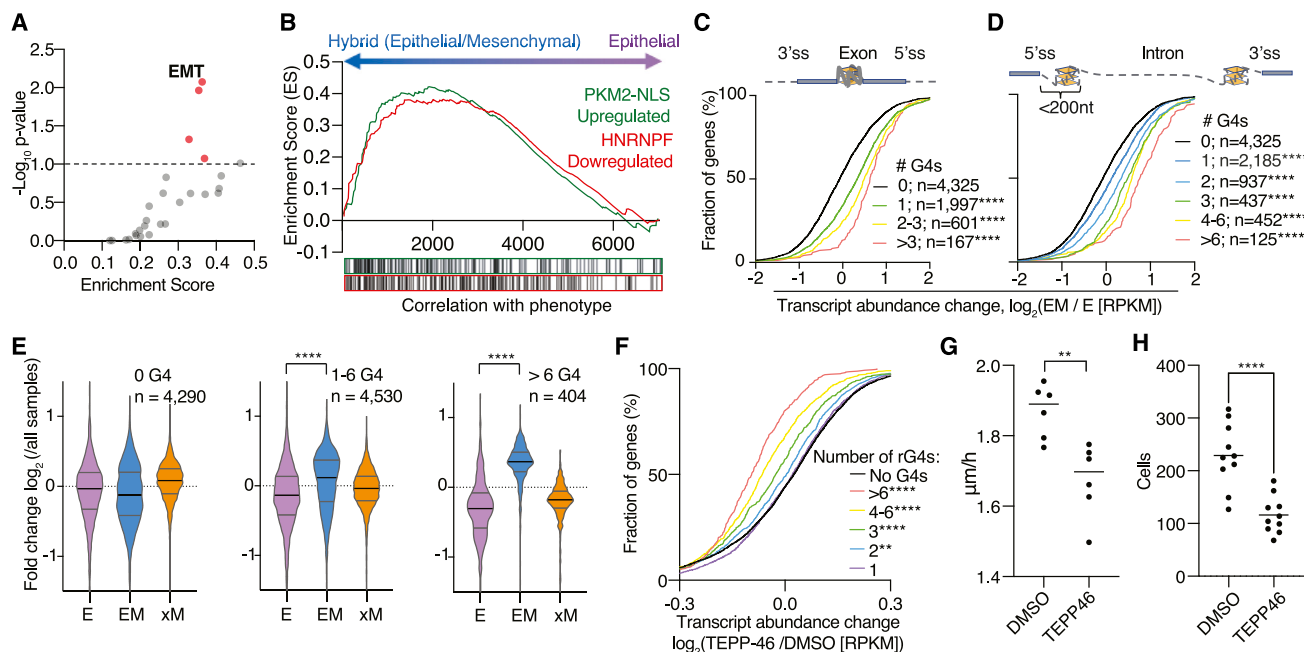


Figure 6. The rG4ome is associated with aggressive cancer phenotype

(A) Scatterplot of hallmark gene sets enriched among genes upregulated by FH-PKM2-NLS expression ($-\log_{10} p$ value versus enrichment score). (B) Gene set enrichment analysis of transcripts upregulated by nuclear PKM2 (determined in HEK293) in HMLER cells as they transition from a less tumorigenic E-state to the highly aggressive hybrid E/M state. (C and D) mRNA expression changes comparing cells in the aggressive E/M state with the basic E-state. Transcripts were binned according to the number of canonical rG4 motifs (GGG(N₁₋₇)₃-GGG) in their exons (C) in their introns within 200 nt of the splicing sites (D). (E) Violin plots of the relative expression of transcripts with 0 (left), 1–6 (center), or more than 6 (right) PRG4S in their exons or introns near a splicing site for E, EM, and xM cells. (F) mRNA expression changes upon treatment of MDA-MB-231 cells with 5 μ M TEPP-46. Transcripts were binned according to the number of canonical rG4 motifs (GGG(N₁₋₇)₃-GGG) in their exons or within 200 nt of the 5' or 3' splice site. (G) Dot plot of cell front velocity in wound healing assay. (H) Dot plot of the proportion of invading cells per tested area of Matrigel-coated transwell filters in invasion assays. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$; two-tailed t test.

also reduced nuclear PKM2 levels and attenuated aggressive phenotypes in a TNBC model. We therefore treated MDA-MB-231 TNBC cells with TEPP-46 to reduce nuclear PKM2 levels and profiled rG4ome levels by RNA-seq. Indeed, TEPP-46-treated cells showed a reduction of the rG4ome as well as decreased migratory and invasive properties (Figures 6F–6H). This confirms that nuclear PKM2 localization plays a critical role in shaping the transcriptome of highly aggressive cancer phenotypes via rG4 interactions. Consequently, we reasoned that cell lines that are engineered to exclude PKM2 from the nucleus would show reduced tumorigenicity in xenograft models. Therefore, we used CRISPR-Cas9-mediated gene editing to introduce an FH tag on the N terminus and an NES on the C terminus of the PKM1/2 proteins in MDA-MB-231 cells. We obtained a set of cells homozygous (PKM-NES-FH/PKM-NES-FH) for the edits (Figure 7A). Cell lines heterozygous for the introduction of an FH tag on the PKM1/2 N terminus (PKM-FH/WT) served as controls. We again assessed the metabolic activity of purified PKM2 from these cell lines and ruled out that introduction of the FH tag or the NES interfered with PKM2 tetramerization or metabolic activity (Figures 7B and S7C). PKM-NES-FH/PKM-NES-FH cells showed decrease migration in a scratch

assay (Figure 7C). We used these cell lines for xenograft experiments and injected 10^7 cells into immunocompromised mice and followed tumor growth. Tumors from homozygous PKM-NES-FH/PKM-NES-FH cells grew slower (Figures 7D and 7E) compared with those from heterozygous PKM-FH/WT or parental MDA-MB-231 cells. Exclusion of PKM2 from the nucleus also allowed mice to survive longer in the xenograft experiments (Figure 7F). RNA-seq analysis of extracted tumor tissue from the primary tumor showed that as in our other experiments, exclusion of PKM2 from the nucleus resulted in reduced expression of the rG4ome (Figure 7G). Since MDA-MB-231 xenografts result in robust metastases, we analyzed lungs for presence and size of metastases once the primary tumors reached a size of $\sim 1,600$ mm³. We examined lung sections across five animals each, xenografted with parental MDA-MB-231, MDA-MB-231^{PKM-FH/WT}, and MDA-MB-231^{PKM-NES-FH/PKM-NES-FH}, respectively. Mice implanted with homozygous cells showed a striking, more than 5-fold decrease in the relative size of metastases (Figure 7H), consistent with our observation that the rG4ome upregulated by PKM2 is enriched for genes from cell migratory pathways (Figure 6A). Taken together, our xenograft experiments supported the data we collected, showing that (1) translocation

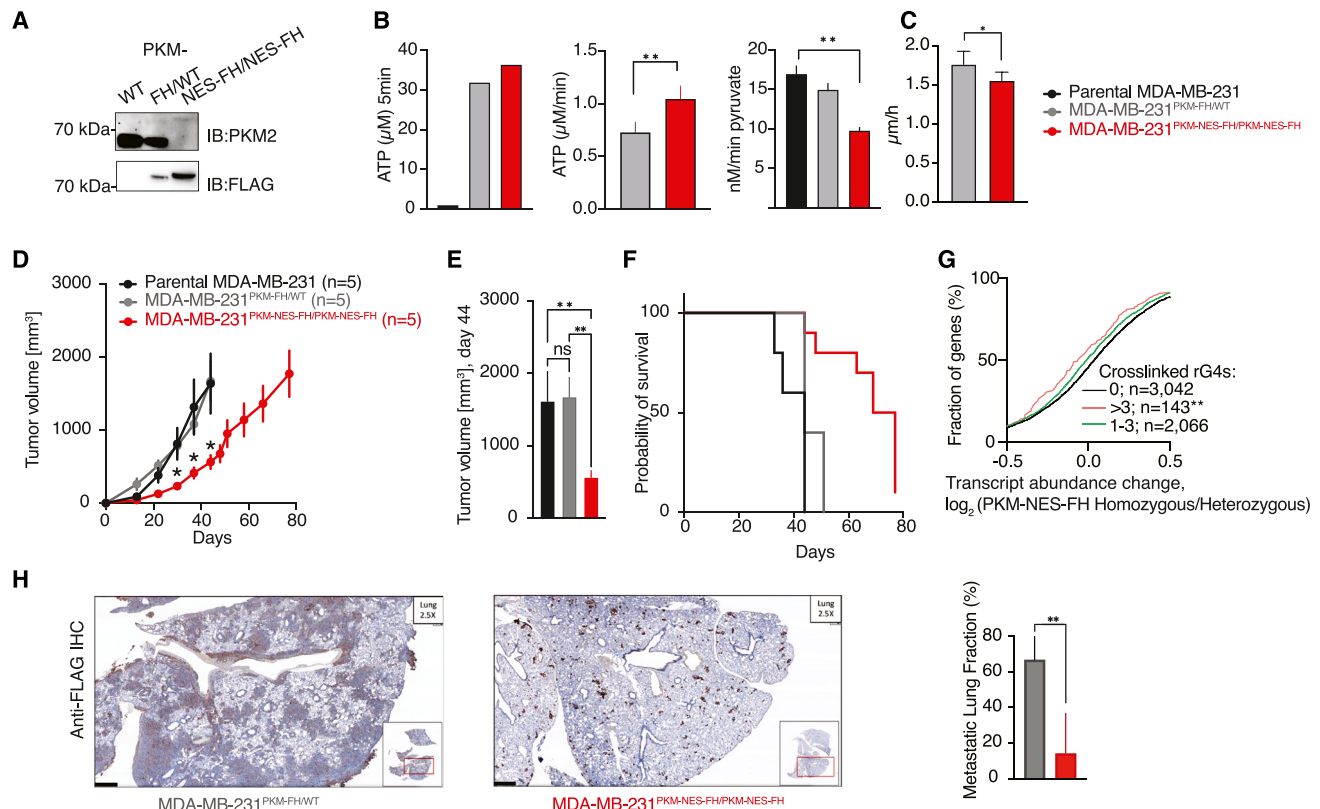


Figure 7. Nuclear PKM2 promotes lung metastasis

(A) Immunoblot analysis showing expression of PKM endogenously tagged with either FH or NES-FH using anti-FLAG or anti-PKM2 antibodies. (B) ATP production in the presence of ADP and pyruvate (left, endpoint; middle, kinetics) from proteins purified by anti-FLAG immunoprecipitation (IP) from parental MDA-MB-231^{PKM-FH/WT} and MDA-MB-231^{PKM-NES-FH/WT} cells. (Right) Pyruvate kinase assay using extracts from parental cells. (C) Dot plot of cell front velocity in wound healing assay of MDA-MB-231^{PKM-FH/WT} and MDA-MB-231^{PKM-NES-FH/WT} cells. (D) Exclusion of PKM2 from the nucleus decreases tumor size. Primary tumor size of xenografts in NOD scid gamma (NSG) mice subcutaneously injected in the right flank with 10^7 cells of either parental MDA-MB-231 (black line), heterozygous PKM-FH/WT (gray line), or homozygous PKM-FH-NES cells. (E) Primary tumor size from (D) 44 days after injection. (F) Kaplan-Meier survival curves for the three groups from (D). (G) mRNA expression changes comparing gene expression in the primary tumor of xenografts of MDA-MB-231 cells homo- or heterozygous for FH-NES-PKM expression. Transcripts for cumulative distribution function were binned by the number of PRG4Ss identified by fPAR-CLIP from parental MDA-MB-231 cells. (H) Representative images of FLAG immunohistochemistry (IHC) staining in lung sections from metastases from xenografts of PKM-FH/WT heterozygous or PKM-NES-FH/WT homozygous MDA-MB-231 cells (left and middle); scale bars, 1 mm. Mean values of fraction of metastasis in lungs (by volume, right).

of PKM2 to the nucleus results in the upregulation of the rG4ome and (2) that upregulation of the rG4ome results in an increase tumor growth and metastatic burden.

DISCUSSION

PKM2 as rG4-interacting RBP

Our study shows that, in addition to its well-characterized role in the energy metabolism, PKM2 can bind rG4 in the nucleus, establishing it among the >20 metabolic enzymes known to bind RNA.²⁵ The effects of RNA-binding to metabolic enzymes have been poorly characterized but include (1) modulation of their enzymatic activity, as seen for GAPDH²⁶ or ENO1,²⁷ (2) feedback control of expression of mRNAs encoding components of the same metabolic pathways, as shown for IRP1 binding to ferritin mRNAs to regulate iron metabolism,²⁸ or (3) use of RNA as scaffold

to support the formation of either larger assemblies of the metabolic enzymes to form a “metabolon,”⁷⁷ or of heteromeric complexes, analogous to the proposed role of RNA in the formation of the PRC2 complex.^{78,79} Translocation of PKM2 to the nucleus removes it from its normal role in glycolysis and allows it to post-transcriptionally regulate the expression of hundreds of mRNAs. Our results suggest this mechanism could contribute to the large-scale rewiring of gene expression programs during metastasis, but also in the context of other phenotypic transitions, such as in development, where nuclear PKM2 is required for T cell differentiation.⁸⁰

Mechanistically, our data suggest that PKM2 affects the efficiency of co-transcriptional mRNA processing feeding back on transcription itself,⁸¹ considering (1) that it acts in the nucleus, (2) that intronic and exonic rG4 sites close to splice sites confer regulation, (3) that intronic rG4 sites found more than 200 nt

removed from splice sites are occupied but do not affect RNA levels, (4) that we find no evidence of systematic splicing pattern changes, and (5) that importantly, we find increases in transcriptional elongation rates. Intriguingly, the Tollervey group found that RNA secondary structures forming close to elongating RNA polymerases promote transcription and prevent backtracking,⁸² providing an elegant mechanistic rationale for the function of rG4 structures—which are among the most stable RNA secondary structures—in mRNA expression. We predict that the interplay and competition of RBPs stabilizing or melting rG4 structures during transcription will profoundly affect the transcriptional output from rG4-containing pre-mRNA transcripts. PKM2 and HNRNPF may just be paradigmatic examples of such factors, highlighting the need to completely map the complement of RBPs interacting with PRG4Ss in the nucleus in different tissues and during development.

Folded rG4 are gene regulatory elements in pre-mRNA

Our study sheds light on the widely debated function of rG4s and highlights their role as *cis*-regulatory elements controlling gene expression. Previous systems-wide studies found that rG4 structures are globally unfolded in the cytoplasm due to the action of a highly redundant machinery of RBPs and helicases that either prevent their formation or unwind them.^{40,51,52,83} Knockout of these factors and concomitant rG4 formation in the cytoplasm is typically detrimental to the cell, e.g., by interfering with translation initiation or elongation,^{34,52} or rG4 self-aggregation, resulting in the activation of stress response pathways.^{51,84} However, rG4 sequences in the vertebrate transcriptome are deeply conserved,^{54,83} suggesting their regulatory role. We show that PKM2 allows co-transcriptional folding of rG4 elements, leading to transcriptome-wide changes in gene expression. Considering that rG4-containing transcripts, including PKM2 targets, are enriched among transcripts required for the migratory phenotype in cancer cells, the rG4 elements may be considered as a clear example of a “regulon”^{85,86} that allows for the coordinated regulation of these transcripts. PKM2 may cooperate or compete with other nuclear rG4-binding RBPs for the regulation of these mRNAs. Consistent with this idea, we showed that overexpression of HNRNPF, which binds unfolded rG4 sequences, has the opposite effect of PKM2. Intriguingly, HNRNPF expression suppresses the EMT,⁷⁶ in contrast to nuclear PKM2, which promotes the expression of EMT associated genes.

PKM2-rG4 interaction as a therapeutic intervention point

While PKM2 expression and activity have been well documented in many types of cancer, investigations with PKM2 knockout mice have challenged the role of PKM2 in tumorigenesis. PKM2 isoform-specific deletion showed no effect or enhanced tumor formation in breast,⁸⁷ colorectal,⁸⁸ and liver cancer.^{89,90} Nevertheless, it is evident that pharmacological activation or inhibition of PKM2 might serve as a platform to target tumor cells by altering their metabolism.^{10,87} In addition, in a mouse embryonic fibroblast oncogenic system, PKM1 expression is consistent with enhanced tumor growth, driven by a cell-autonomous mechanism of glycolytic control that is not documented for

PKM2.⁹¹ These studies have uncovered a nuanced role for pyruvate kinase activity via both PKM1/2. Furthermore, other work suggests that cell type and context may be important for the role of PKM2 controlling cell proliferation and tumor growth.^{7,89,92} Our study interrogates the effects of non-metabolic function of nuclear PKM2 in the context of tumor progression without connecting PKM2 function to tumor etiology. We restricted PKM2 from the nucleus (PKM2-NES) to resemble acute changes related to PKM2 RNA-binding function without disrupting PKM2 metabolic activity. Accordingly, we did not observe changes in the enzymatic activity of our PKM2-NES but rather showed a change in the rG4ome across multiple cell lines. Importantly, our genetically modified PKM2-NES cells show striking changes with respect to their capacity of metastatic seeding in an aggressive breast cancer xenograft model. Our results point to nuclear regulation of rG4 transcripts as a possible mechanism for the integration of metabolic and signaling inputs that could control metastasis.

We propose that the PKM2-rG4 interaction may be the key to understanding the tumorigenic activity of nuclear PKM2. Disruption of this interaction could represent a therapeutic vulnerability to target migratory cell phenotypes, which coordinate tumorigenesis and metastasis. PKM2-rG4 interactions could be manipulated in multiple ways, e.g., (1) by influencing PKM splicing patterns using oligoribonucleotide inhibitors suppressing exon 10 inclusion, as prominently used to correct SMN splicing,⁹³ (2) prevention of PKM2 nuclear translocation, e.g., stabilization of the cytoplasmic form,¹⁰ or interference with PKM2 phosphorylation,¹³ or (3) structure-guided targeting of the, yet to be determined, PKM2-rG4 interaction surface.

Limitations of the study

A major limitation of our study is the subtlety of the observed differences in the RNA-seq and ChIP-seq upon manipulation of nuclear PKM2 levels. On average, we saw changes of ~10% on mRNA levels for the top 200 PKM2 targets upon depletion of PKM2 from the nucleus in HEK293 levels, complicating validation of downstream protein expression changes. Nevertheless, the consistency between the ChIP-seq and RNA-seq data supports our interpretation that the regulation occurs at the transcriptional level. Furthermore, we found that *in vitro*, PKM2 binds DNA as well as RNA G4 structures *in vitro*. While G4s in promoter elements and in the template strand of gene bodies do not show any sensitivity to changes of nuclear PKM2 levels, we cannot fully rule out that DNA G4 elements in the coding strand and other G4s, such as those at telomeres, may be regulated by PKM2. Finally, while the correlation of rG4ome levels with poor patient prognosis in various cancers from TCGA is suggestive, further statistical analysis and experimental evidence need to be collected before the rG4ome can be unambiguously nominated as a general cancer driver.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.molcel.2024.07.025>.

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AUTHOR CONTRIBUTIONS

Conceptualization, D.G.A., M.A., J.R., and M.H.; methodology, D.G.A., M.A., K.A.G., M.I.U., Q.M., J.S., J.E.J., and X.W.; computational analysis, D.G.A. and A.H.P.; investigation, D.G.A., M.A., and K.A.G.; writing – original draft, D.G.A., M.A., M.H., and J.R.; writing – review and editing, D.G.A., M.A., J.S., K.A.G., I.B., and J.R.; funding acquisition, M.H., J.R., I.B., and A.D.H.; resources, M.H., J.R., I.B., and A.D.H.; supervision, M.H. and J.R.

DECLARATION OF INTERESTS

J.R. is on the scientific advisory board and has an equity interest in Pearl Bio. J.R. is a co-founder and has an equity interest in Kapis Biosciences.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
<i>Anti-FLAG M2 Magnetic Beads</i>	Sigma-Aldrich	M8823
Anti-FLAG M2 antibody	Sigma-Aldrich	F3165
Alexa Fluor 488 Secondary Antibody	Thermo Fisher Scientific	A-11034
Alexa Fluor 594 Secondary Antibody	Thermo Fisher Scientific	A-11032
Anti-HNRNP Antibody		hpa069667
Anti-PKM2 (D78A4)	Cell Signaling	4053
Anti-PKM	Abcam	ab150377
Anti-RNA polymerase II CTD repeat YSPTSPS antibody - ChIP Grade	Abcam	ab26721
Anti-RNA polymerase II CTD repeat YSPTSPS (phospho S2) antibody	Abcam	(ab5095)
Anti-RNA polymerase II CTD repeat YSPTSPS (phospho S5) antibody	Abcam	(ab5131)
GAPDH	ProteinTech	10494-1-AP
Bacterial and virus strains		
E. coli BL21	NEB	C2527H
E. coli C321.ΔA	Addgene	48998
E. coli 5-alpha	NEB	C2987H
Chemicals, peptides, and recombinant proteins		
4-Thiouridine	BIOSYNTH	NT06186
ATP Solution (100 mM)	Thermo	R0441
Blasticidin S	Gibco	A1113903
Blasticidin S HCl	Gibco	A1113903
Chloroform:Isoamyl alcohol 24:1	Sigma	C0549-1PT
cOmplete Protease Inhibitor Cocktail	Roche	1.1697E+10
Deoxynucleotide (dNTP) Solution Mix	NEB	10966034
DMEM, high glucose	Gibco	11965092
Dynabeads Protein G for Immunoprecipitation	Invitrogen	10004D
EMEM	ATCC	30-2003
Fetal Bovine Serum	GeminiBio	100-800
FLAG Peptide	Sigma	F3290
Gateway BP Clonase II Enzyme mix	Invitrogen	11789020
Gateway LR Clonase II Enzyme mix	Invitrogen	11791020
GlycoBlue Coprecipitant	Thermo Fisher Scientific	AM9515
HisPur™ Ni-NTA Magnetic Beads	Thermo Scientific	88831
Hygromycin B	Gibco	10687010
Insulin solution from bovine pancreas	Sigma	I0516
IPTG	MP Biomedicals	02102101-CF
Lipofectamine 3000	Invitrogen	L3000001
Lipofectamine RNAiMAX	Invitrogen	13778075
Opti-MEM I Reduced Serum Medium	Gibco	31985070
PageRuler Plus, Prestained Protein Ladder	Thermo Fisher Scientific	26620
Penicillin-Streptomycin	Gibco	15140163

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Platinum Taq DNA polymerase	Thermo Fisher Scientific	10966034
Proteinase K	Millipore Sigma	3115879001
Puromycin Dihydrochloride	Gibco	A1113803
Quick CIP	NEB	M0525S
RNase I	Thermo Fischer Scientific	AM2294
RNase T1	Thermo Fischer Scientific	EN0542
Rnl1 T4 RNA ligase	Thermo Fischer Scientific	EL0021
SUPERase-In RNase inhibitor	Thermo Fisher Scientific	AM2694
SuperScript IV	Thermo Fisher Scientific	18090010
T4 PNK	NEB	M0201S

Critical commercial assays

EnzyChrom Pyruvate Kinase Assay Kit	Bio Assay Systems	EPRK-100
EnzyLight ATP Assay kit	Bio Assay Systems	EATP-100

Deposited data

RNA-seq	This study	GSE179817
PARCLIP	This study	GSE179817
RBNS	This study	GSE179817
CHIP-seq	This study	GSE179817
Normalized TCGA expression data	TCGA	GSE62944
Hybrid E/M state breast cancer cells	Kröger et al. ⁷⁵	GSE119149

Experimental models: Cell lines

HEK293 T-Rex Flp-In	Thermo Fisher Scientific	R78007
MCF-7	ATCC	HTB-22
MDA-MB-231	ATCC	HTB-26
293T/17 [HEK 293T/17]	ATCC	CRL-11268

Experimental models: Organisms/strains

NOD scid gamma (NSG) mice	The Jackson Laboratory	005557
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Oligonucleotides

DsPKM 3'UTR	IDT	hs.Ri.PKM.13.3
DsSCRAMBLE	IDT	51-01-14-03

Plasmids

pOG44 Flp-Recombinase Expression Vector	Invitrogen	V600520
pLKO.1	Addgene	8453
pFRT_TO_DESTFLAGHA	Addgene	26361
pMDLg/pRRE	Addgene	12251
pRSV-Rev	Addgene	12253
pMD2.G	Addgene	12259
pET-28a-hPKM1	Addgene	44241
pET-28a-hPKM2	Addgene	44242
pCRT7 NT Topo	Addgene	73446
SepOTSλ	Addgene	68292

Software and algorithms

ImageJ	Fiji	SCR_002285
GraphPad Prism	GraphPad	https://www.graphpad.com/
ChimeraX	Pettersen et al. ⁹⁴	http://plato.cgl.ucsf.edu/chimerax/download.html
Samtools Version: 1.9	Li et al. ⁹⁵	https://samtools.sourceforge.net/
PyMOL	Schrödinger	https://www.pymol.org

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Cutadapt v2.4	Martin ⁹⁶	https://github.com/marcelm/cutadapt
STAR v2.7.2b	Dobin et al. ⁹⁷	https://github.com/alexdobin/STAR
Other		
Slide-A-Lyzer™ Dialysis Cassette, 10K MWCO	Thermo Fisher Scientific	66380
Pierce™ Protein Concentrators PES	Thermo Fisher Scientific	88513
Ni-charged MagBeads	GenScript	L00295

RESOURCE AVAILABILITY

Lead contact

Requests for resources and reagents and further information should be directed to and will be fulfilled by the lead contact, Dr. Markus Hafner (markus.hafner@nih.gov).

Materials availability

All unique/stable reagents generated in this study are available from the [lead contact](#) with a completed Materials Transfer Agreement.

Data and code availability

- Data produced within this study are publicly available as of the date of publication from the Gene Expression Omnibus (GEO) under the accession number GSE179817. This paper analyzes existing, publicly available data. The accession numbers for the datasets are listed in the [key resources table](#).
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mouse model

Animal studies were performed in accordance with protocols approved by the Animal Care and Use Committee (ACUC) of the National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS). Six-week-old female NOD scid gamma (NSG) mice were obtained from The Jackson Laboratory and housed at NIAMS animal facilities with standard chow and water provided ad libitum. Mice were subcutaneously injected (23G needle; Med-Plus, Inc., Edison, NJ) with 10^7 cells of either parental MDA-MB231 ($n = 5$), heterozygous FH-NES-PKM/WT ($n = 5$), or homozygous FH-NES-PKM ($n = 10$) cells in 50% Matrigel (Corning, Corning, NY)/50% PBS in a total volume of 100 microliters. Tumor size was measured once per week. Study endpoints were body weight loss exceeding 20% or tumor size exceeding 15 mm in its largest diameter.

Cell culture

HEK293 T-Rex Flp-In cells (Thermo Fisher Scientific, Cat# R78007) were cultured in DMEM medium (Gibco) supplemented with 10% (v/v) fetal bovine serum (FBS), 100 μ g/ml zeocin, and 10 μ g/ml blasticidin (Gibco). MCF-7 cells were obtained from ATCC (Cat# HTB-22) and cultured in the recommended EMEM medium (ATCC) supplemented with 10% (v/v) FBS and 0.01 mg/ml bovine insulin (Sigma). MDA-MB-231 cells were obtained from ATCC (Cat# HTB-26) and cultured in DMEM-high glucose (Gibco) supplemented with 10% (v/v) FBS.

METHOD DETAILS

Plasmid transfection and mRNA knockdown

pFRT/TO-FLAG/HA-PKM2 and pFRT/TO-FLAG/HA-PKM2-NLS and -NES plasmids were generated as described previously.⁹⁸ Plasmids were transfected in cells together with pOG44 plasmid expressing Flp-recombinase (Invitrogen) using Lipofectamine 2000 transfection reagent (Invitrogen) according to manufacturer's instructions to create stable HEK293 T-Rex Flp-In cells that conditionally express (doxycycline inducible) either wild-type Flag-HA-tagged PKM2 or Flag-HA-tagged PKM2-NES/-NLS fusions. Stable cell lines were maintained under selection using 10 μ g/ml blasticidin and 50 μ g/ml hygromycin B (Gibco).

Knockdown of PKM2 in MCF-7 cell lines was performed by lentiviral transduction. pLKO.1 (Addgene #84530) and 3rd generation lentiviral vectors were obtained from Addgene (#12251, #12253, #12259). shRNAs targeting the PKM 3' UTR and control scramble shRNA were designed and cloned into pLKO.1 plasmid. Lentiviral and expression plasmids were co-transfected (1:1 ratio) into

HEK293T/17 cells (ATCC) using Lipofectamine 3000 transfection reagent (Invitrogen) according to manufacturer's instructions. The viral containing supernatant were collected 24 and 48 hours after transfection, filtered, and used to infect MCF-7 cells in the presence of 10 μ g/ml polybrene. Infected cells were cultured for 9 days with 2 μ g/ml puromycin (Gibco). Silencing of endogenous PKM in cells expressing PKM2-NES was performed using siRNA targeting 3'UTR of PKM (IDT hs.Ri.PKM.13.3) or scramble control (IDT, 51-01-14-03) using LipofectamineTM RNAiMAX Transfection Reagent.

shPKM DNA oligomers

5'-CCGCAACGCTTGTAGAACTCACTCCTCGAGGAGTGAGTTCTACAAGCGTTGTTTTG
5'-AATTCAAAAACAACGCTTGTAGAACTCACTCCTCGAGGAGTGAGTTCTACAAGCGTTG

Scramble-shRNA DNA oligomers

5'-CCGGCCTAAGGTTAAGTCGCCCTCGCTCGAGCGAGGGCGACTTAACCTTAGGTTTTG
5'-AATTCAAAAACCTAAGGTTAAGTCGCCCTCGCTCGAGCGAGGGCGACTTAACCTTAGG

fPAR-CLIP

Fluorescent Photoactivatable ribonucleoside-enhanced crosslinking and immunoprecipitation (fPAR-CLIP) was performed as described in Anastasakis et al.⁶⁷ with several modifications. For Doxycycline induced expression in HEK293 cells, 10 $\times$ 15 cm plates at 70% confluency cells conditionally expressing the wild-type PKM2/HNRNPF were induced with doxycycline for 24 hours followed by 100 μ M 4-thiouridine treatment for 12 hours and UV crosslinked at 365 nm (300 mJ/cm²). The cytoplasm was extracted by resuspending cells in a modified hypotonic lysis buffer (10 mM Tris-HCl pH 7.5, 50 mM NaCl, 3 mM MgCl₂, 0.1% NP-40, 10% glycerol) for 2 min. The nuclear pellet was washed twice with the same buffer and lysed in NP-40 buffer, followed by sonication. Proteins were immunoprecipitated using the anti-Flag M2 magnetic beads (Sigma). For fPAR-CLIPs of FH-PKM2 from HEK293 cells, after FLAG-IP, the RNPs were dephosphorylated and phosphorylated, and a 5' fluorescent DNA adapter was modified with a 5' IRDye[®] 800CW and four degenerate nucleotides that serve as unique molecular identifiers at the 3' end (IR800-GTTCAGAGTTCTACAGTCCGACGATCrNrNrNrN). After SDS-PAGE analysis the fluorescent bands corresponding to the RNP-adapter were isolated, shredded, and treated with proteinase K as described in Anastasakis et al.⁶⁷ 3' adapter (NNTAGTCGTGGAATTCTCGGGTGCCAAGG for cytoplasm and NNAGTGTCTGGAATTCTCGGGTGCCAAGG for nuclear) ligation was then performed on the recovered RNA footprint. For all other fPAR-CLIPs a 3' fluorescent adapter (5'-rAppNNTGACTGTGGAATTCTCGGGT(AF647)GCCAAGG-(AF647)) was first ligated following IP and dephosphorylation. After SDS-PAGE analysis and proteinase K digestion phosphorylation was performed and a 5' adapter was ligated (5'(aminolinker)-GTTCAGAGTTCTACAGTCCGACGATCrNrNrNrN). cDNA was synthesized using RT primer. A Low cycle PCR was performed using Platinum Taq DNA polymerase (Thermo Fisher Scientific, 10966034), a short 5' PCR (CTTCAGAGTTCTACAGTCCGACGA) primer and the RT primer according to manufacturer's instruction. PCR products were size selected, for a range of 78-98 bp, on a 3% agarose gel (3% Pippin gel cassettes, Sage Science, CSD3010). A pilot PCR using as template 25% of the first low cycle and short primer PCR with 5' long PCR primer: AATGATACGGCGACCACCGAGATCTACACGTTTCAGAGTTCTACAGTCCGA and 3' RNA PCR primer index: CAAGCAGAAGACGGCATACGAGAT-(6 or 8 base index)-GTGACTGGAGTTCCTTGGCACCCGAGAATTCCA. After 8 cycles, 15 μ l of product was removed every two cycles (8, 10, 12, ..., 18 cycles). Products were visualized on a 2.5% agarose gel. The lowest cycle number with visible amount of product was chosen for a final large-scale PCR (30 μ l of size selected in 300 μ l total) with the same primers. The PCR products were cleaned and concentrated using DNA Clean & Concentrator kit (Zymo Research, D4013) and size selected (~140-170 bp) to remove primers and adapter-adapter byproducts using a 3% agarose gel (3% Pippin Prep). Quality of the final library was assessed using TapeStation system (Agilent). Sequencing was performed on an Illumina HiSeq 3000 or NovaSeq 6000 platform.

Recombinant protein expression and purification

To express recombinant PKM1 and PKM2 proteins, pET-28a-hPKM1 (Addgene #44241) and pET-28a-hPKM2 (Addgene #44242) were transformed into *E. coli* BL21(DE3) cells. 0.4 L of LB cultures were induced with 0.5 mM IPTG at an OD₆₀₀ of 0.5 and incubated for an additional 5–7 hours at RT. Cell pellets were resuspended in 8 ml lysis buffer (300 mM NaCl, 50mM Tris-HCl pH 7.5, and 1% NP40), 20 mM imidazole, 1 mM DTT, and protease inhibitor cocktail (cOmplete, Roche). 0.5 ml of HisPurTM Ni-NTA Magnetic Beads (Thermo Fisher Scientific), equilibrated in the same buffer, were incubated with the lysate for 1 hour at 4°C under rotation and washed three times with 10 ml wash buffer (600 mM NaCl, 50mM Tris-HCl pH 7.5, 1% NP40, 30 mM Imidazole). Proteins were eluted with 1.5 ml elution buffer (300 mM NaCl, 50 mM Tris-HCl pH 7.5, 1% NP40 and 180 mM imidazole). Purified proteins were dialyzed into storage buffer (50 mM Tris-HCl pH 7.5, 300 mM NaCl) using Slide-A-LyzerTM Dialysis Cassette, 10K MWCO, 3 ml (Thermo Fisher Scientific, Cat# 66380) and concentrated using Protein Concentrators PES, 10K MWCO, 0.5 ml (Thermo Fisher Scientific, Cat# 88513).

Phosphoprotein synthesis and purification

Plasmids and strains

DNA sequences encoding the full-length N-term 6xHis-tagged Human PKM2 including a S37STOP(TAG) mutation was assembled into pCRT7/NT-TOPO bacterial expression plasmid (Addgene #73446) at NdeI/SacI cloning sites, respectively. pCRT7/NT-TOPO (Addgene #68306) was transformed into genomically recoded *E. coli* (C321. Δ A) cells along with SepOTS λ plasmid for phosphoprotein production (Addgene #68292).^{99,100}

Expression and purification

Expression and purification of the full-length PKM2pS37 protein was performed as described before with modifications.¹⁰¹ In brief, expression was induced with 0.2% arabinose at an OD₆₀₀ of 0.8–0.9 for 20 hours at 30°C. Bacterial cells were lysed in 20 mM Tris-HCl pH 7.2, 500 mM NaCl, 1 mM Tris (2-carboxyethyl) phosphine (TCEP), 50 mM NaF, 1 mM NaVO₄ buffer supplemented with 1 mg/ml lysozyme (Sigma), and protease inhibitor (cOmplete, Roche). Recombinant protein was purified on a HisTrap HP column (GE Healthcare) using continuous imidazole gradient (10–500 mM), followed by size exclusion chromatography on a Superdex 200 10/300 GL (GE Healthcare) with 20 mM Tris-HCl pH 7.2, 100 mM NaCl buffer containing 0.5 mM DTT, and 5% glycerol. Purified proteins were buffer exchanged into 50 mM Tris-HCl pH 7.4, 150 mM NaCl, and 20% glycerol for storage at -20°C.

Immunofluorescence analysis

Seven to 8 × 10⁴ cells were seeded in ibiTreat chamber slides (Ibidi). After 20–24 hours, cells were fixed with 4% PFA for 15 min at room temperature, washed three times with 1× DPBS, permeabilized with 0.1% Triton X-100 for 15 min and blocked with 10% goat serum (constituted in 0.1% Triton X-100) at room temperature. Cells were stained overnight at 4°C with primary antibodies against PKM2pS37 (rabbit polyclonal, custom-made) and total PKM2 (Cell Signaling Technology, Cat# 4053) a 1:200 and 1:100 dilution in blocking solution, respectively, or FLAG antibody (Sigma-Aldrich, Cat# F3165) (1:200 dilution) followed by incubation with secondary antibodies (1:300 dilution) conjugated with Alexa Fluor 488 (Thermo Fisher Scientific, Cat# A-11034) or Alexa Fluor 594 (Thermo Fisher Scientific, Cat# A-11032) and Hoechst 33342 (Invitrogen) labeling. All samples were imaged by confocal microscopy on a Leica Laser Scanning SP8 Microscope (20x or 63x oil objective) at room temperature using the same settings. Images were analyzed using ImageJ software (Fiji, RRID: SCR_002285). Fluorescence intensity was determined using manual masking and identical brightness and contrast parameters.

Electromobility shift assay

For Electromobility shift assays (EMSA) 6xHis-tagged proteins were incubated in increasing concentrations with 15 nM of Cy-5 fluorescent oligonucleotide in binding buffer (25 mM Tris-HCl pH 7.5, 100 mM KCl, 3 mM MgCl₂, 0.01% tween, 1 mg/ml BSA, and 1 mM DTT) adjusted with water to a final volume of 40 μl. After incubation for 30 min at room temperature 10 μl of 50% glycerol was added and reactions were separated on a 0.6% agarose gel (running buffer 1× TBE, 25 mM KCl) at 80 V for 90 min at room temperature.

Bind-n-seq assay

Bind-n-seq analysis was performed as described previously with a few modifications.⁵⁵ Five micrograms of 6x His-tagged PKM2 protein was bound to 25 μl Ni-Charged MagBeads (GenScript, L00295), washed, and incubated with 1 μM synthetic, completely randomized 29-nt long oligoribonucleotides (IDT) in binding buffer (25 mM Tris-HCl pH 7.5, 100 mM KCl, 3 mM MgCl₂, 0.01% Tween, 1 mg/ml BSA, and 1 mM DTT) for 30 min at room temperature. Beads were washed once with binding buffer and RNA and proteins eluted with 10 mM Tris pH 7.0, 1 mM EDTA, 1% SDS. Eluted RNA was purified using the Oligo Clean & Concentrator kit according to manufacturer's instructions (Zymo Research) and a 3' adapter was ligated using T4 RNA Ligase 2, truncated (NEB) following manufacturer's instructions. The reaction was cleaned and a 5' adapter was ligated using T4 RNA Ligase (Ambion, Thermo Fisher Scientific), followed by reverse transcription using SuperScript IV Reverse Transcriptase and RT primer (GCCTTGGCACCCGAGAATTCCA) and PCR amplified using Platinum™ Taq DNA Polymerase and Illumina RNA PCR primers.

RNA Sequencing

RNA from cells was isolated using the Direct-zol RNA Miniprep Kit (Zymo Research, Cat# R2050) according to the manufacturer's instructions. One microgram total RNA was Ribosomal RNA depleted using the NEBNext® rRNA Depletion Kit and cDNA libraries were prepared using the NEBNext® Ultra™ Directional RNA Library Prep Kit for Illumina® (NEB). cDNA libraries were sequenced on the Illumina HiSeq 3000 HiSeq X or NovaSeq 6000 platform. Reads were aligned to human genome version hg38 using STAR (star/2.7.2b).⁹⁷ Cufflinks was used for differential expression.¹⁰² All gene set files were obtained from GSEA website (www.broadinstitute.org/gsea). GSEA was performed using the GSEA v4.1.0 software. GO enrichment analysis was performed using the ShinyGO v0.66 tool.¹⁰³

ChIP-seq

ChIP-seq was performed following abcam protocols. Two by 15 cm plates of HEK293 T-Rex FH-PKM2-NLS (Treated with Doxycycline or DMSO for 72 hours) were crosslinked with 0.75% (Alfa Aesar 43368) for 10 min followed by 5 min treatment with 125 mM Glycine. Cells were washed and collected with PBS and lysed with 0.5 ml of lysis buffer (50 mM HEPES-KOH pH7.5, 140 mM NaCl, 1 mM EDTA pH8, 1% Triton X-100, 0.1% Sodium Deoxycholate, 0.1% SDS, and protease inhibitor cocktail [cOmplete, Roche]). After sonication using a bath sonicator (Bioruptor, diagenode) for 14 min (medium amplitude, 30 s on 30 s off) at 4°C. Next, 0.25 ml of lysate was diluted in RIPA buffer (50 mM Tris-HCl pH 8, 150 mM NaCl, 2 mM EDTA pH 8, 1% NP-40, 0.5% Sodium Deoxycholate, 0.1% SDS, and protease inhibitor cocktail) to a volume of 2 ml and IP was performed using 5 μg of anti-RNA polymerase II CTD repeat YSPTSPS antibody (Abcam, ab26721) or Anti-RNA polymerase II CTD repeat YSPTSPS (phosphor-S2) (Abcam, ab5095) for 1 hour at 4°C. 30 μl of Protein G Dynabeads were equilibrated in the same buffer and added to the samples for 16 hours at 4°C under rotation. Beads were washed once with the following buffers. Low Salt Wash Buffer (0.1% SDS, 1% Triton X-100, 2 mM

EDTA, 20 mM Tris-HCl pH 8.0, and 150 mM NaCl), High Salt Wash Buffer (0.1% SDS, 1% Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 8.0, and 500 mM NaCl), LiCl Wash Buffer (0.25 M LiCl, 1% NP-40, 1% Sodium Deoxycholate, 1 mM EDTA, and 10 mM Tris-HCl pH 8.0). Elution was performed with 120 μ l elution buffer at 30°C for 15 min with elution buffer (1% SDS and 100mM NaHCO₃). Four and eight-tenths microliters of 5 M NaCl and 2 μ l Rnase A/T1 (Thermo EN0551) were added to the eluent and incubated while shaking at 65°C overnight. Two microliters proteinase K (20 mg/mL) and incubate while shaking at 60°C for 1 h. Phenol/chloroform extraction was performed and the sample was cleaned and concentrated using the DNA Clean & Concentrator-5 kit Zymo Research, D013. Library for illumine sequencing was prepared using the NEBNext® Ultra™ II DNA Library Prep Kit for Illumina.

Matrigel invasion assays

Invasion assays were carried out as described previously using Biocoat Matrigel Invasion Chambers (BD Biosciences).¹³ Cells were serum starved for 6 hours and seeded on the chamber well inserts (5 × 10⁴ cells/insert). Cells were let to invade for 20–24 hours. Cells that did not invade were removed from the inner surface of the Matrigel chamber using a moistened cotton swab. The remaining, migrated cells were fixed directly on chamber by soaking in 4% PFA. After washing with 1× DPBS, membranes were stained with Hoechst 33342 (Invitrogen) and mounted on slides using ProLong Gold Antifade Mountant (Invitrogen). Imaging was performed using a Fluorescence Microscope (10× objective) and taken images (6–8 fields/membrane) were analyzed with ImageJ software.

Kinase activity assays

Pyruvate kinase reactions was performed under the following conditions: 30 mM Tris pH 7.4, 10 mM MgCl₂, 0.6 mM PEP, and 1.5 mM ADP (final concentrations) with or without the presence of F1,6BP (70 μ M). Reactions were assembled to final volume of 100 μ l with the addition of Kinase-Glo Reagent (Promega) according to manufacturer's instructions. The reactions were then initiated by adding the enzyme alone or in complex with equal amount (22 nM) of folded rG4 oligonucleotide (sequence derived from SLC7A5 3'UTR G4). ATP production was followed by measuring luminescence over time on a SpectraMax i3x plate reader (Molecular Devices). Measurements were converted to enzymatic units (μ mol/min) and specific activities were calculated by normalizing to amount of protein (in mg).

TCGA analysis

Normalized TCGA expression data were downloaded from GEO GSE62944. The processed dataset file "GSM1536837_06_01_15_TCGA_24.tumor_Rsubread_FPKM.txt.gz" was used for further analyses. The genes were classified arbitrarily to G4-High and G4-Low if the number of rG4s were ≥ 4 and ≤ 2 , respectively. Borderline genes (rG4 = 3) were omitted for better classification. Expression values for genes were log₂ transformed. For each of the cancer samples (columns of TCGA dataset) means of expression for G4-High and G4-Low were calculated. Rank for the individual cancer samples (columns of TCGA data) were calculated by the ratio of the above-mentioned means by following formula: rankValue = (mean of G4 High genes) / (mean of G4 Low genes). Higher "rankValue" indicates have high expression values of genes with more than four rG4 per gene. For each of the 24 cancer subtypes a hyperparameter optimization cycle were carried out to find out the maximum survival differences among samples. To do so samples were stratified via percentiles of rankValue, i.e., samples with top 10% rankValue vs rest, top 20% rankValue vs rest, and so on. This systematic approach segregated the samples that has high expression values of genes with more than four rG4s that appeared to show low survivals compared to the rest.

Immunohistochemistry

Slides were first deparaffinized in xylene and then hydrated through graded alcohols. Next, slides were subject to antigen retrieval in a Tris/EDTA buffer (pH 9). Slides were then blocked against endogenous peroxidases. Following blocking, slides were incubated with the primary antibody (Abcam ab205606, DDDDK) at a dilution of 1/1,350 (0.65ug/ml). The antibody was then detected with a goat-anti-rabbit HRP secondary and visualized with DAB (brown). Finally, the slides were counterstained with hematoxylin, dehydrated in graded alcohols, cleared in xylene, and mounted with permanent mounting media. All rinses were performed using either distilled water or TBST.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses was performed using GraphPad Prism software. Bar graphs were presented using mean values with error bars representing SD.